Affine-Invariant Gaussian Gradient Flows: Mode Decomposition and the Role of (ω, τ)

Abstract

We study a two-parameter family of affine-invariant Gaussian gradient flows for the variational problem $\min_{m,C} \mathrm{KL}(\mathcal{N}(m,C) \,\|\, \pi)$. The family is parameterized by (ω, τ): the scalar $\omega > 0$ rescales the covariance dynamics uniformly, while $\tau > -\omega/n$ enters only through a trace-weighting term and therefore acts solely on the volume (trace) mode of the covariance. A linearization near stationarity predicts that the trace-mode rate is $\kappa_{\mathrm{tr}} = 1/[2(\omega + n\tau)]$, so a negative τ accelerates volume-dominated transients while leaving the mean and the traceless shape modes unchanged. We verify these predictions on two targets—an exact Gaussian and a strongly log-concave non-Gaussian target—using an exponential-map discretization that preserves positive-definiteness. The measured time-to-tolerance ratios match the predicted factors ($\approx 2\times$ acceleration under $\tau_- = -\omega/2n$, $\approx 3/2$ slowdown under $\tau_+ = +\omega/2n$) for volume-dominated initializations, and the realized speedup tracks the initial trace-dominance of the perturbation.

1 Setup

Let the variational family be the non-degenerate Gaussians $q_{m,C} = \mathcal{N}(m, C)$ with $m \in \mathbb{R}^n$ and $C \in \mathrm{SPD}(n)$, and let the target be $\pi(\theta) \propto \exp(-V(\theta))$ with V smooth. The objective is the reverse Kullback–Leibler divergence

$$F(m, C) = \mathrm{KL}\big(\mathcal{N}(m, C) \,\|\, \pi\big) = \mathbb{E}_{\theta \sim \mathcal{N}(m,C)}[V(\theta)] - \tfrac{1}{2} \log\det C + \mathrm{const}. \tag{1}$$

Writing $g(m, C) = \mathbb{E}_{q_{m,C}}[\nabla V]$ and $S(m, C) = \mathbb{E}_{q_{m,C}}[\nabla^2 V]$, the first-order stationarity conditions are $g(m_\star, C_\star) = 0$ and $C_\star^{-1} = S(m_\star, C_\star)$.

1.1 Classification of affine-invariant metrics

A Riemannian metric on the Gaussian manifold $\mathcal{A} = \mathbb{R}^n \times \mathrm{SPD}(n)$ is *affine-invariant* if it is preserved by the pushforward of every invertible affine map $\theta \mapsto A\theta + b$.

Theorem 1 (Classification)**.** *Every affine-invariant Riemannian metric on $\mathcal{A}$ has the form*

$$g_a\big((u, X), (v, Y)\big) = u^\top C^{-1} v + \omega \ \mathrm{Tr}(C^{-1} X C^{-1} Y) + \tau \ \mathrm{Tr}(C^{-1} X) \ \mathrm{Tr}(C^{-1} Y), \tag{2}$$

with $a = (m, C)$, $\omega > 0$, and $\tau > -\omega/n$. There is no mean–covariance coupling.

The admissibility condition $\omega + n\tau > 0$ is what keeps the induced flow well-posed; it follows from positive-definiteness of the metric on the trace mode.

1.2 The (ω, τ) family of flows

Taking F as the energy and the metric of Theorem 1 as the geometry, the gradient flow is

$$\dot{m}_t = -C_t\, g(m_t, C_t), \qquad \dot{C}_t = \frac{1}{2\omega}\, C_t^{1/2}\big(- B_t + \alpha_t I\big) C_t^{1/2}, \tag{3}$$

where $B_t = C_t^{1/2} S(m_t, C_t) C_t^{1/2}$ is the Hessian expectation in the whitened coordinates of C_t and

$$\alpha_t = \frac{\omega + \tau \ \mathrm{Tr}(B_t)}{\omega + n\tau} \tag{4}$$

is a scalar trace-weighting term. The parameter ω rescales the entire covariance equation; τ enters *only* through the scalar α_t, and α_t multiplies the identity, so τ acts only on the trace of the covariance residual.

1.3 Exponential-map discretization

With step Δt the implemented update is

$$m_{k+1} = m_k - \Delta t\, C_k\, g(m_k, C_k), \qquad C_{k+1} = C_k^{1/2} \exp\Big(\tfrac{\Delta t}{2\omega}\big(- B_k + \alpha_k I\big)\Big) C_k^{1/2}. \tag{5}$$

Because $-B_k + \alpha_k I$ is symmetric, its matrix exponential is symmetric positive definite, and C_{k+1} is a congruence transform of an SPD matrix; hence $C_{k+1} \in \mathrm{SPD}(n)$ at every step, with no projection. This is the discrete exponential map on the SPD manifold.

2 Trace-mode linearization

The effect of τ is made precise by linearizing near stationarity. Consider an isotropic perturbation of the whitened Hessian, $B = (1+\varepsilon)I$, so that $\mathrm{Tr}(B) = n(1+\varepsilon)$ and

$$\alpha = \frac{\omega + \tau n(1+\varepsilon)}{\omega + n\tau} = 1 + \frac{n\tau}{\omega + n\tau}\, \varepsilon, \qquad \Longrightarrow \qquad -B + \alpha I = -\frac{\omega}{\omega + n\tau}\, \varepsilon\, I. \tag{6}$$

Substituting into (3) near $C = I$ gives a linear decay $\dot{\varepsilon} = -\kappa_{\mathrm{tr}}\, \varepsilon$ for the trace mode with rate

$$\kappa_{\mathrm{tr}} = \frac{1}{2(\omega + n\tau)}. \tag{7}$$

Three consequences follow for the volume/trace mode, all independent of the mean and shape modes:

- $\tau = 0$: rate $\kappa_{\mathrm{tr}} = 1/(2\omega)$;
- $\tau_- = -\omega/2n$: $\omega + n\tau = \omega/2$, so $\kappa_{\mathrm{tr}} = 1/\omega$ — the rate doubles, the convergence time halves;
- $\tau_+ = +\omega/2n$: $\omega + n\tau = 3\omega/2$, so $\kappa_{\mathrm{tr}} = 1/(3\omega)$ — the time scale grows by 3/2.

The mean equation in (3) does not contain τ, and a traceless (shape) residual has $\mathrm{Tr} = 0$, on which α has no leading-order effect. Thus τ can accelerate *only* a trace-dominated transient.

To quantify how trace-dominated a residual is, decompose the covariance residual R_{cov} into trace and traceless parts and define the *trace-dominance ratio*

$$\chi = \frac{\big(\,\mathrm{Tr}\, R_{\mathrm{cov}}\big)^2}{n\, \|R_{\mathrm{cov}}\|_F^2} \in [0, 1], \tag{8}$$

with $\chi \approx 1$ a purely volume residual and $\chi \approx 0$ a purely shape residual. By (7) the realized τ-speedup should be largest when $\chi_0 \approx 1$.

Table 1: Configuration of the two (ω, τ) grids. The Gaussian target uses exact expectations; the log-concave target uses a fixed Monte Carlo sample set with a numerically computed reference optimum $F_\star$.

quantity	Gaussian target	log-concave target
dimension n	$\{2, 5, 10\}$	5
ω grid	$\{0.125, 0.25, 0.5, 1, 2\}$	$\{0.25, 0.5, 1\}$
τ per ω	$\{-\omega/2n,\ 0,\ +\omega/2n\}$	$\{-\omega/2n,\ 0,\ +\omega/2n\}$
step Δt	0.02	0.005
horizon T	20	40
coupling ρ	–	5
features $m = 4n$	–	20
dynamics samples K	exact	4096
reference objective $F_\star$	0 (exact)	3.9809

3 Experiments

Two targets are used. The **Gaussian** target $V(\theta) = \frac{1}{2}\|\theta\|^2$, $\pi = \mathcal{N}(0, I)$, has exact expectations $g(m, C) = m$, $S(m, C) = I$, equilibrium $(m_\star, C_\star) = (0, I)$, and admits an exact diagonal covariance update in the eigenbasis of C. The **log-concave** target

$$V_\rho(x) = \tfrac{1}{2}\|x\|^2 + \frac{\rho}{m}\sum_{\ell=1}^{m} \log\cosh(a_\ell^\top x), \qquad I \preceq \nabla^2 V_\rho \preceq (1+\rho)I, \tag{9}$$

with $m = 4n$ fixed unit directions a_ℓ, is 1-strongly convex and $(1+\rho)$-smooth. Its expectations are estimated with a fixed Monte Carlo sample set (common random numbers across all $(\omega, \tau, \text{init})$ runs), and the reference optimum $a_\star = (m_\star, C_\star)$ is computed by L-BFGS-B on a fixed-sample estimate of F over a Cholesky factor.

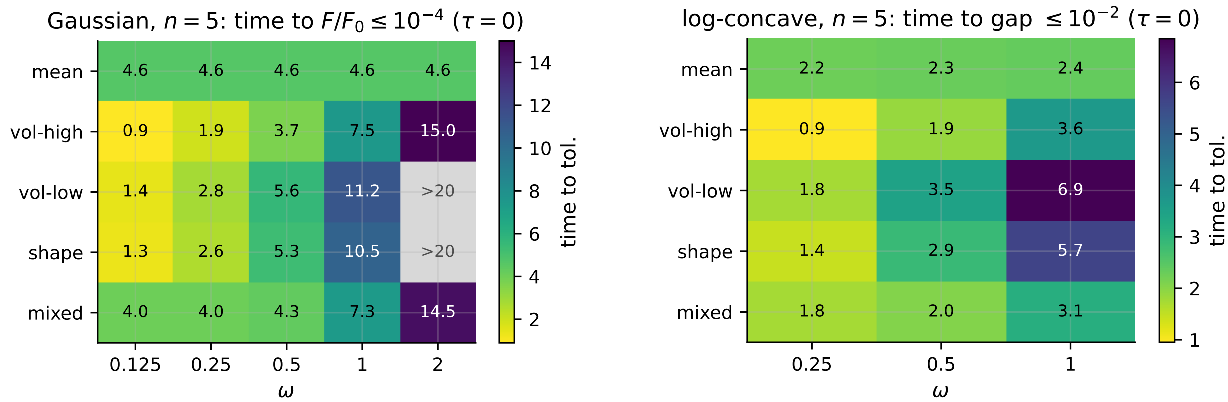
Five initializations isolate the convergence modes: *mean* (mean displaced, $C_0 = C_\star$), *vol-high*/*vol-low* (isotropic $C_0 = 4C_\star$ / $0.25\,C_\star$, $\chi_0 \approx 1$), *shape* (traceless, $\det C_0 = \det C_\star$, $\chi_0 \approx 0$), and *mixed* (all modes). For each ω the three values $\tau \in \{-\omega/2n, 0, +\omega/2n\}$ are run. The grid metadata is given in Table 1.

4 Results

Time scale set by ω. Figure 1 shows the time-to-tolerance at $\tau = 0$ as a function of ω and initialization, for both targets. The mean mode is essentially flat in ω (its equation does not involve ω or τ), whereas the covariance-dominated modes slow monotonically as ω grows, consistent with the $\kappa_{\mathrm{tr}} \propto 1/\omega$ scaling of (7).

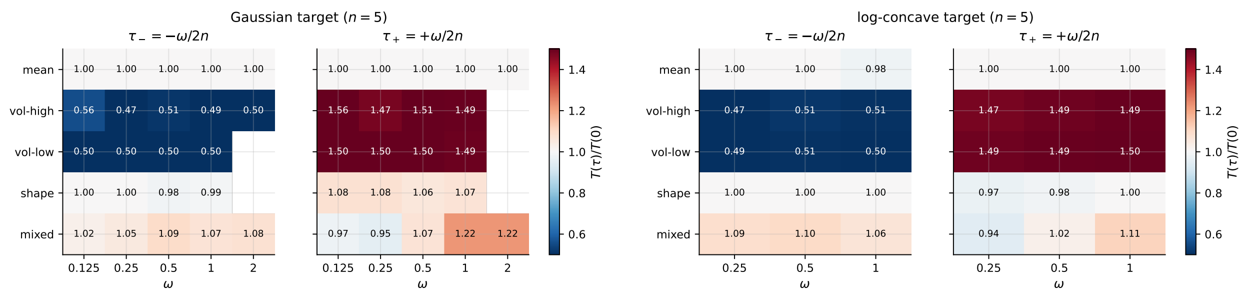
τ accelerates only the volume mode. Figure 2 reports the time-to-tolerance ratio $T(\tau)/T(0)$. For both targets and uniformly in ω, the two volume initializations accelerate by a factor close to 2 under τ_- (ratios 0.47–0.51) and slow by a factor close to 3/2 under τ_+ (ratios 1.47–1.51), matching the predicted trace-mode rates of Section 2. The mean and shape modes stay within a few percent of 1, confirming that τ leaves them unaffected. Table 2 lists the ratios at $\omega = 0.5$.

The speedup tracks trace dominance. Figure 3 plots the realized τ_- speedup against the initial trace-dominance χ_0 of (8) for the log-concave target. The near-2× acceleration ($T(\tau_-)/T(0) \approx$



(a) Gaussian target. **(b)** log-concave target.

Figure 1: Time to reach the convergence tolerance at $\tau = 0$, by initialization (rows) and ω (columns). The mean mode is independent of ω; the volume and shape modes slow as ω increases. Cells that do not reach the tolerance within the horizon are masked and labeled $> T$.



(a) Gaussian target. **(b)** log-concave target.

Figure 2: Speedup ratio $T(\tau)/T(0)$ (diverging scale centered at 1). Blue < 1 is faster than $\tau = 0$, red > 1 is slower. Only the volume modes move: ≈ 0.5 under τ_- and ≈ 1.5 under τ_+, across all ω.

0.5) is realized only for $\chi_0 \approx 1$; as $\chi_0 \to 0$ the ratio returns to 1. This is the mechanism behind the mode selectivity: τ rescales the trace channel, so its benefit is proportional to how much of the limiting error lives in that channel.

5 Discussion

Within the affine-invariant family, no single (ω, τ) is uniformly optimal; the parameters trade off cleanly separated channels.

- ω sets the covariance time scale uniformly. Smaller ω accelerates every covariance mode but does not affect the mean; the mean channel is governed by its own (parameter-free) equation.
- τ acts only on the volume mode, with rate $\kappa_{\mathrm{tr}} = 1/[2(\omega + n\tau)]$. A negative τ (subject to $\omega + n\tau > 0$) accelerates volume-dominated transients—up to the $2\times$ realized here at $\tau_- = -\omega/2n$—and is otherwise neutral.

The practical recommendation is therefore conditional on the dominant error mode: when the limiting error is volume/trace dominated (χ_0 near 1), choosing $\tau < 0$ as negative as admissibility

Table 2: Time-to-tolerance and τ-speedup ratios at $\omega = 0.5$ (Gaussian tolerance 10^{-4}, log-concave 10^{-2}). $T(0)$ is in flow-time units.

	Gaussian (10^{-4})			log-concave (10^{-2})		
initialization	$T(0)$	$T(\tau_-)/T_0$	$T(\tau_+)/T_0$	$T(0)$	$T(\tau_-)/T_0$	$T(\tau_+)/T_0$
mean	4.6	1.00	1.00	2.3	1.00	1.00
vol-high	3.7	0.51	1.51	1.9	0.51	1.49
vol-low	5.6	0.50	1.50	3.5	0.51	1.49
shape	5.3	0.98	1.06	2.9	1.00	0.98
mixed	4.3	1.09	1.07	2.0	1.10	1.02

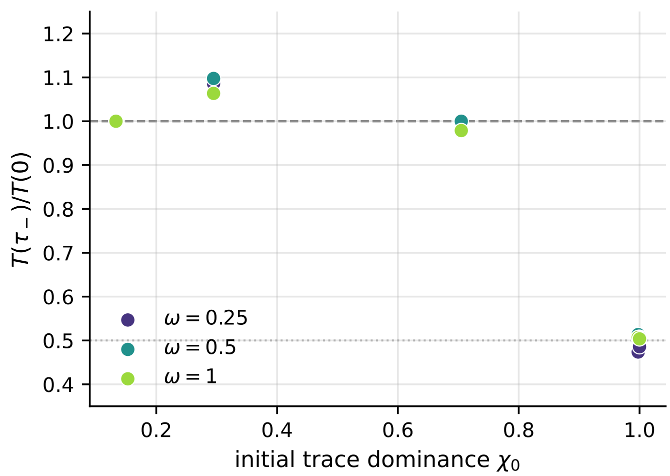


Figure 3: Realized τ_- speedup versus initial trace-dominance χ_0 (log-concave target, all ω). The acceleration is realized only for trace-dominated initializations ($\chi_0 \approx 1$); shape- and mean-dominated initializations (χ_0 small) show no net speedup.

allows shortens the convergence time; when the limiting error is shape- or mean-dominated, τ should be left at 0 and only ω tuned.

Limitations. The experiments cover $n \le 10$ (Gaussian) and $n = 5$ (log-concave) with the grids of Table 1. The log-concave expectations are Monte Carlo estimates with a fixed sample set; the normalized objective gap has a numerical floor near 10^{-4} for that run, so the speedup ratios are read at the 10^{-2} tolerance, which is well above the floor. The reported factors are empirical time-to-tolerance ratios consistent with the linearized rates; they are not a global convergence proof.